

ANIRUDH GOYAL

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EDUCATION

- **Indian Institute of Technology Delhi** July 2023 - Present
B.Tech. & M.Tech in Chemical Engineering - CGPA: 7.823/10.0 Delhi, India
- **Guru Nanak Khalsa Senior Secondary School** March 2023
Class XII CBSE - Percentage: 91.60% Chandigarh, India
- **Little Flower Convent School** March 2021
Class X ICSE - Percentage: 92.60% Panchkula, Haryana, India

SKILLS

- **Programming Languages:** C, C++, Java, Python, SQL, MATLAB, L^AT_EX
- **Frameworks/Libraries:** Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, XGBoost, PyTorch, FastAPI, LangChain
- **Tools/Platforms:** Git, GitHub, PostgreSQL, Jupyter, Google Colab, Ollama, VS Code, CAD
- **Concepts:** Machine Learning, Deep Learning, Time Series Forecasting, RAG, Prompt Engineering, RLHF, Computer Vision
- **Coursework:** Data Structures & Algorithms, Probability and Stochastic Processes, Linear Algebra, Calculus, Differential Equations, Numerical Methods, Macroeconomics

WORK EXPERIENCE

- **Giesecke & Devrient**  May 2025 - July 2025
Machine Learning Intern – Forecasting Systems Gurgaon, Haryana
 - Built an **XGBoost**-based sales forecasting model with monthly **incremental** and **biannual full retraining**; improved **RMSE** by up to **15%** via lag features, rolling window stats, and modular preprocessing.
 - Designed **PostgreSQL** schema for predictions; integrated with **FastAPI** backend to serve real-time forecasts and KPI metrics on an interactive dashboard.
 - Developed a **RAG**-based chatbot using **Ollama** and **Meta LLaMA**, enabling context-aware Q&A on business datasets through prompt engineering.
- **Outlier / Scale AI**  March 2025 - May 2025
AI Data Specialist (Independent Contractor) Remote – San Francisco, USA
 - Applied **RLHF** and **prompt engineering** to enhance model performance on truthfulness, safety, and reasoning tasks across diverse domains.
 - Designed complex **Chain-of-Thought (CoT)** prompts to improve AI reasoning and step-by-step problem decomposition; conducted **Quality Assurance** reviews on prompts and outputs.
 - Analyzed AI-generated responses against **Ground Truth Final Answers (GTFA)**; contributed to projects *Gallon Hamster* and *Green Wizards*.

PROJECTS

- **Helmet Safety Detection using YOLOv8 & Faster R-CNN** January 2025 - March 2025
Computer Vision – Self Project 
 - Trained and optimized **YOLOv8** and **Faster R-CNN** on the **Kaggle Helmet Detection Challenge** dataset; implemented end-to-end pipelines in **PyTorch**.
 - Compared models on **accuracy**, **inference speed**, and real-time applicability, identifying trade-offs for safety-critical use cases.
- **Predictive Modelling of Bio-Diesel Production** October 2024 - December 2024
Prof. Jayati Sarkar 
 - Compiled face-centered **CCD datasets (29+ runs)** from published literature; benchmarked **Linear Regression**, **Random Forest**, and **AdaBoost** models.
 - Identified **AdaBoost** as best performer (**R²=0.97**, **MSE ≈ 0.25**); recommended optimal reaction conditions to achieve **95%+ yield**.
- **Optimal Strategy Development for Competitive Game Theory** August 2024
Prof. Dharmaraja Selvamuthu 
 - Developed a **Dynamic Programming** approach with **Monte Carlo** simulations to compute probabilities and optimize across 3 distinct play styles within a **20-second** runtime constraint.

ACHIEVEMENTS & CERTIFICATIONS

- **JEE Advanced 2023:** Secured **All India Rank 4216** (top **0.4%**); improved from JEE Mains AIR **9310**. 2023
- **Course Star Certificate:** Awarded by IIT Delhi for solving a star-level problem in *Intro. to Computer Science (C)*. Dec 2023
- **Coursera:** Completed *Supervised ML – Regression & Classification* by **Stanford Online & DeepLearning.AI**. Nov 2024

LEADERSHIP & CO-CURRICULAR ACTIVITIES

- **Executive**, Alumni Association, IIT Delhi – Organized Convocation, UPSC Panel Talk, AGM. Jul 2024 - May 2025
- **Activity Head**, Rendezvous, IIT Delhi – Led logistics for *Blitzkrieg*, coordinating participants & comms. Jan 2024 - Jan 2025
- **Invigilator**, Office of Career Services, IIT Delhi – Invigilated internship and placement processes. Aug 2024 - Present